

DAXHIEL PERUGORRIA RUCIEL

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EDUCATION

University of Texas at Austin – Austin, TX

May 2029

Bachelor of Science in Computer Science

Coursework: Data Structures and Algorithms, Computer Organization and Architecture, Discrete Mathematics, Linear Algebra, Probability and Statistics

EXPERIENCE

Getfertilizerwiser | St. Edward's University – Austin, TX

2026 - Current

Fullstack Developer

- **Built** a full-stack Next.js 16 (App Router) website for a non-profit environmental campaign using TailwindCSS v4, custom design tokens, and deployed on Vercel.
- **Designed and implemented** a multi-page marketing site with a cohesive brand system, custom typography, CSS animations, and reusable component library.
- **Integrated third-party services** including Formspree for serverless form handling and Cloudinary for media delivery.

Austin Spanish First – Austin, TX

2020 - Current

Co-Director, Technical Team

- **Direct a 10+ member technical team**, delegating tasks and coordinating efforts to deliver seamless audio/visual experiences for 400+ weekly attendees.
- **Redesigned** audio and live-streaming infrastructure, improving broadcast quality and growing online viewership by 50%.

PROJECTS

Recommendation Engine Evaluation Model | *Datahack 2026*

- **Architected a causal Transformer skip-prediction** model for sequential listening behavior by converting each session into a token stream (user attributes + track semantics + context + labels), enabling next-step prediction of the final skip label and reaching 87.07% validation skip accuracy during training.
- **Scaled representation learning** to 1,066,723 tracks by generating 3-level hierarchical “semantic” track IDs via MiniBatchKMeans clustering (32×32×32) on standardized audio features, then encoding each track with clustered IDs + binned popularity/skip-rate signals to improve generalization on long-tail content.

Kalshi Arbitrage Bot | *Python*

- **Built a Python-based arbitrage engine** that integrates with the Kalshi API to scan prediction markets, validate contract data, and algorithmically detect risk-free “bundle long” and “bundle short” opportunities across mutually exclusive outcomes.
- **Designed a modular architecture** with dedicated components for market analysis, opportunity batching, trade execution with risk checks, and an interactive Rich/Questionary-powered CLI for discovering, inspecting, and executing arbitrage strategies in real time.

Fake-or-Not | *Python, FastAPI, React*

- **Built an end-to-end fake-news classification pipeline** using Jupyter notebooks for data exploration and feature engineering with data libraries, and exported the trained model as an ONNX artifact for portable inference.
- **Implemented a production-ready model** serving backend exposing a REST API for real-time predictions, including request validation, model versioning, and batch prediction endpoints to support both online and offline workflows.
- **Developed a responsive single-page frontend** in React (Vite) with fetch-based integration to the REST /predict endpoint, styled with modern CSS and maintained with ESLint for code quality.

TECHNICAL SKILLS

- **Frameworks & Tools:** Pytorch, FastAPI, Next.js, React, Vercel, Supabase, Express, NEAT, SQL
- **Languages:** Python, Java, Javascript, C/C++
- **Work Eligibility:** Eligible to work in the U.S. with no restrictions